Post-Repair Quality Acceptance — All Assets

Universal QA Verification Procedure — All Asset Types and Fault Modes
Applies after: SOP_001, SOP_002, SOP_003, SOP_004, SOP_005
Bearing Types: All | Est. Duration: 1–2 h | Crew: 1 | Priority: MANDATORY
SOP_006 | v1.1 | Revised: 2026-03-20

1. Purpose and Scope

This procedure defines the universal post-repair quality acceptance check performed after any maintenance intervention on plant assets AST_MTR_001, AST_MTR_002, AST_PMP_001, AST_PMP_002, AST_CON_001, and AST_GBX_001. The QA check must be performed at two time points after every repair: at 1 hour post-restart and at 24 hours post-restart. A work order cannot be closed as 'successful' unless both QA checks are documented. This SOP is read by both the Monitoring Agent (to confirm QA pass/fail thresholds) and the Learning Agent (to capture confirmed post-repair baselines for future reference).

2. Post-Repair Acceptance Thresholds by Asset

Asset ID	Bearing IDs	Vib RMS ≤ (mm/s)	Temp ≤ (°C)	Kurtosis ≤	BPFO ≤ (×baseline)
AST_MTR_001	BRG_001 (DE), BRG_002 (NDE)	2.5	60°C at 1h | 55°C at 24h	2.8	1.2×
AST_MTR_002	BRG_003 (DE), BRG_004 (NDE)	2.2	55°C	2.5	1.2×
AST_PMP_001	BRG_005 (DE), BRG_006 (NDE)	3.5	65°C	3.5	1.2×
AST_PMP_002	BRG_007 (DE), BRG_008 (NDE)	3.0	60°C	3.0	1.2×
AST_CON_001	BRG_009 (DRIVE), BRG_010 (ND)	2.8	58°C	2.8	1.2×
AST_GBX_001	BRG_011 (INPUT), BRG_012 (OUT)	3.5	65°C at 1h | 62°C at 24h	2.8	1.2×

3. Step-by-Step QA Procedure

1	Wait for machine to reach stable operating speed and rated load before taking the 1-hour reading.
2	Record vib_rms_mm_s from portable vibration meter at the bearing housing (or from historian for 1-hour average).
3	Record temp_c from infrared thermometer at bearing housing outer surface (or from historian).
4	Record kurtosis from historian (1-hour rolling average from latest telemetry record).
5	Record signal_quality_score from latest telemetry record for each active channel.
6	Compare all readings against Section 2 thresholds for the specific asset.
7	If ALL criteria pass at 1 hour: annotate work order with 1-hour QA PASS and readings.
8	If ANY criterion fails at 1 hour: STOP. Do not proceed to 24-hour check. Investigate and re-assess.
9	At 24 hours post-restart: repeat steps 2 through 6 using historian data.
10	If ALL criteria pass at 24 hours: annotate work order with 24-hour QA PASS. Work order may be closed.
11	Record post-repair baseline: vib_rms, temp_c, kurtosis as the new reference for this bearing.
12	Update bearing_master.current_health_score to reflect post-repair healthy state.

13	For signal-quality repairs (SOP_004): confirm zero burst dropouts on CH_31B for 15 consecutive readings.
14	Close work order and submit QA record to Learning Agent for case closure.

4. Confirmed Post-Repair Baselines from Historical Cases

Reference values below are confirmed outcomes from closed cases. Use these as expected healthy-state benchmarks when assessing QA results:

Case	Asset	Bearing	Post-Repair Vib (mm/s)	Post-Repair Temp (°C)	Outcome
CASE_001	AST_MTR_001	BRG_001	1.8	55	PASS — bearing replacement
CASE_002	AST_PMP_001	BRG_005	2.2	60	PASS — lubrication service
CASE_003	AST_CON_001	BRG_009	2.1	54	PASS — sensor repair
CASE_005	AST_MTR_002	BRG_003	1.7	51	PASS — preventive check
CASE_006	AST_PMP_002	BRG_007	2.0	57	PASS — vibration inspection
CASE_009	AST_PMP_001	BRG_006	2.1	58	PASS — preventive lube

5. Related Documents

- SOP_001 — Outer Race Bearing Replacement, Motor
- SOP_002 — Preventive Maintenance, Motor
- SOP_003 — Lubrication Service, Pump
- SOP_004 — Sensor Repair, Conveyor
- SOP_005 — Outer Race Bearing Replacement, Gearbox